

Name: Solutions                     / 25

30 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 25 points possible.

1. (5 points) Consider the **telescoping** series  $\sum_{n=1}^{\infty} \left( \frac{1}{n} - \frac{1}{n+2} \right)$ .

(a) Find a simplified formula for  $S_k$ , the  $k$ th partial sum of the series.

$$S_k = \left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{4} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{k-2} - \frac{1}{k}\right) + \left(\frac{1}{k-1} - \frac{1}{k+1}\right) + \left(\frac{1}{k} - \frac{1}{k+2}\right)$$

$$S_k = \frac{3}{2} - \frac{1}{k+1} - \frac{1}{k+2}$$

(b) Use your answer from part (a) to determine the sum of the series.

$$\lim_{k \rightarrow \infty} \left( \frac{3}{2} - \frac{1}{k+1} - \frac{1}{k+2} \right) = \boxed{\frac{3}{2}} = \sum_{n=1}^{\infty} \left( \frac{1}{n} - \frac{1}{n+2} \right)$$

2. (8 points) For each **geometric** series below, determine whether the series converges and **justify** your conclusion. If the series converges, determine the sum.

$$(a) \sum_{n=1}^{\infty} \frac{(-4)^{n-1}}{\pi 5^{n-1}} = \sum_{n=1}^{\infty} \left( \frac{1}{\pi} \right) \left( \frac{-4}{5} \right)^{n-1} = \frac{a}{1-r} = \frac{\frac{1}{\pi}}{1 - \left( \frac{-4}{5} \right)} = \frac{1}{\pi} \left( \frac{1}{\frac{9}{5}} \right) = \boxed{\frac{5}{9\pi}}$$

$$\text{So } a = \frac{1}{\pi}, r = \frac{-4}{5}$$

$|r| < 1$ . So the series converges

$$(b) \sum_{n=1}^{\infty} \frac{2^{2n}}{3^n} = \sum_{n=1}^{\infty} \frac{(2^2)^n}{3^n} = \sum_{n=1}^{\infty} \left( \frac{4}{3} \right)^n$$

Since  $r = \frac{4}{3} > 1$ , the series diverges

3. (8 points) For each series below, determine whether the series converges and **justify** your conclusion. Your justification must include the test or series you are using to draw your conclusion.

(a)  $\sum_{n=1}^{\infty} \frac{n}{10n+17}$

Divergence Test

$$a_n = \frac{n}{10n+17}$$

$$\lim_{n \rightarrow \infty} \frac{n}{10n+17} = \frac{1}{10} \neq 0$$

→ So, by the Divergence Test,  
the series diverges

(b)  $\sum_{n=1}^{\infty} \frac{1}{n^{\sqrt{5}}}$

This is a p-series with  $p = \sqrt{5} > 1$ .  
Thus, the series converges

4. (4 points) Apply the Integral Test to determine if the series  $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln(n)}}$  converges or diverges. (You do not have to show that the Integral Test applies; it does.)

$$\int_2^{\infty} \frac{(\ln x)^{-1/2}}{x} dx = \lim_{t \rightarrow \infty} \int_2^t \frac{(\ln x)^{-1/2}}{x} dx = \lim_{t \rightarrow \infty} \left[ 2(\ln x)^{1/2} \right]_2^t$$

$$= \lim_{t \rightarrow \infty} (2\sqrt{\ln t} - 2\sqrt{\ln 2}) = \infty \quad (\text{as } t \rightarrow \infty, \ln(t) \rightarrow \infty)$$

Since the integral diverges, the series diverges.